

Chapter 1

An introduction to Statistics

Outline

1.3 Populations and Samples

1.4 Three Case Studies That Illustrate Sampling and Statistical Inference (Selecting a Random Sample)

1.3 Populations and Samples

Data

- **Data:** facts and figures from which conclusions can be drawn
- **Data set:** the data that are collected for a particular study
- **Elements:** may be people, objects, events, or other entries
- **Variable:** any characteristic of an element
- **Measurement:** A way to assign a value of a variable to the element
- **Example:** A bank might measure the time it takes for a credit card-holder's bill to be paid to the nearest day.

1.3.1 Variable and Data

- The variable is said to be **quantitative**: Measurements that represent quantities are numbers (for example, “how much” or “how many”). For example, **annual starting salary** is quantitative, **age and number of children** is also quantitative.
- The variable is said to be **qualitative** or **categorical**: Measurements that represent quantities fall into several categories. For example, **a person’s gender**, the **make of an automobile** and whether **a person who purchases a product is satisfied** with the product are qualitative.

Two types of qualitative variables:

- **Nominative**

Unranked categorization

Example: gender, car colour

- **Ordinal**

Rank-order categories

Ranks are relative to each other

Example: low (1), moderate (2), high (3), or extremely high (5) risk

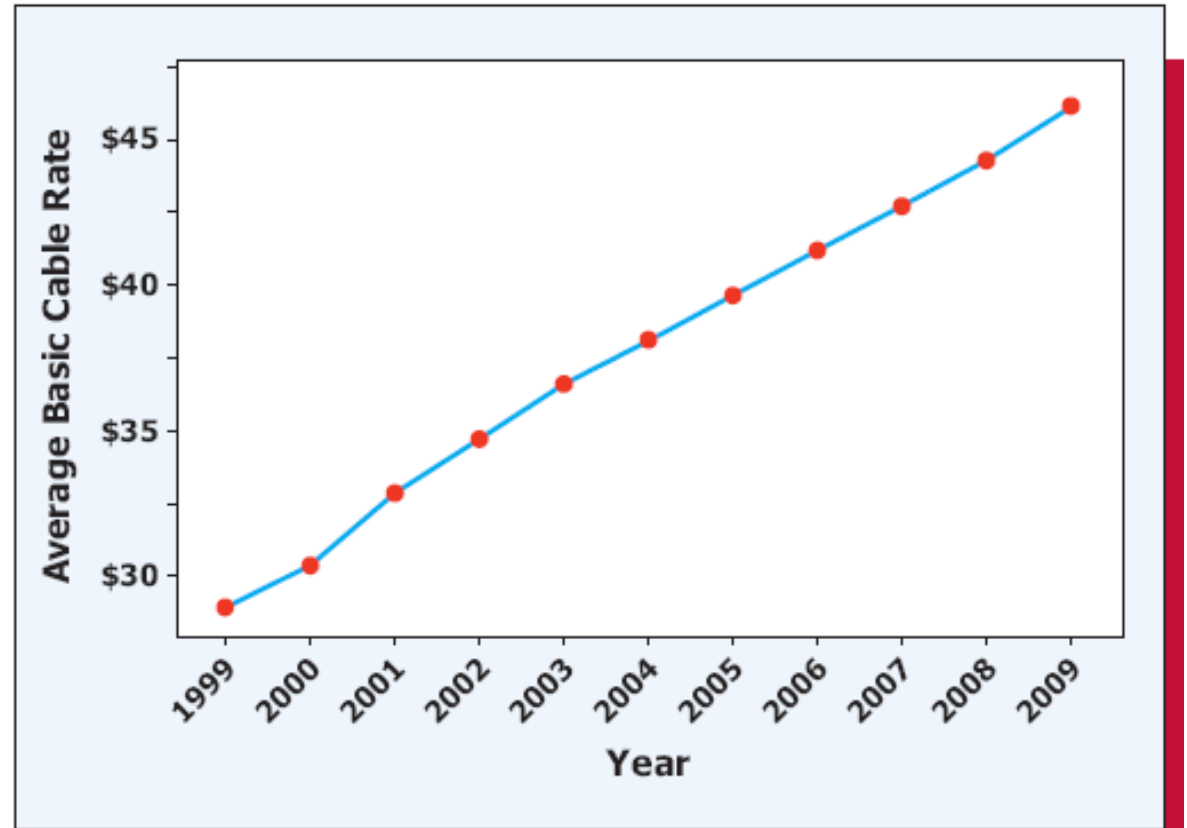
Cross-Sectional Data

- **Cross-sectional data:** Data collected at the same or approximately the same point in time.
- **Time series data:** data collected over different time periods.

Time Series Data

Year	Average Basic Cable Rate
1999	\$ 28.92
2000	30.37
2001	32.87
2002	34.71
2003	36.59
2004	38.14
2005	39.63
2006	41.17
2007	42.72
2008	44.28
2009	46.13

Source: U.S. Energy Information Administration,
<http://www.eia.gov/>



1.3.2 Populations, Samples, Census

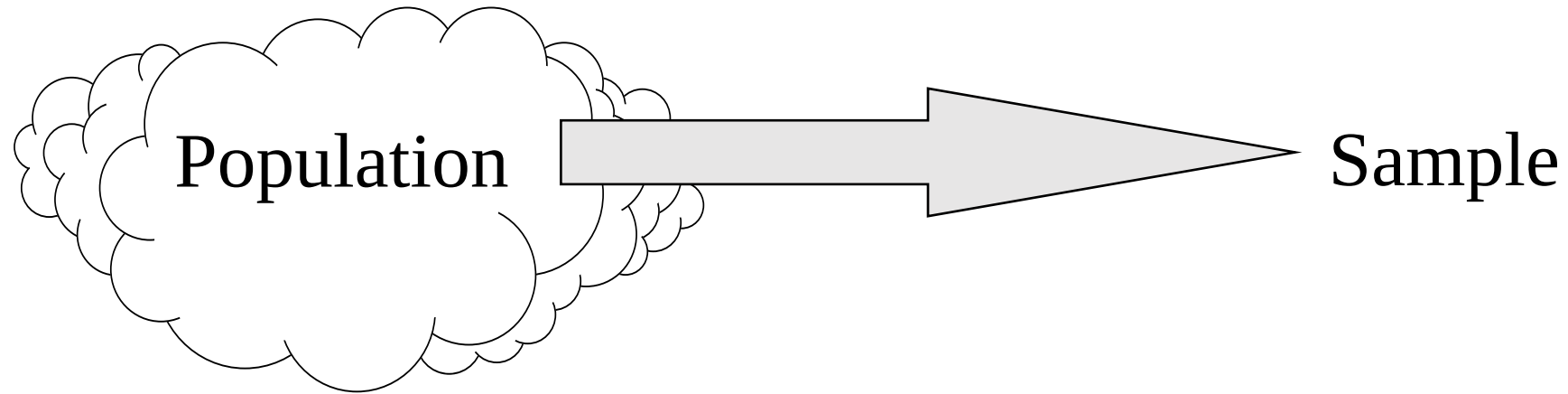
Population: The set of all elements about which we are interested and wish to draw conclusions (people, objects or events)

Examples:

- All of the last year's graduates of HIT's Master of Business Administration program.
- All Lincoln Town Cars that were produced last year.
- All accounts receivable invoices accumulated last year by The Procter & Gamble Company.
- All fire reported last month to the Tulsa, Oklahoma, fire department.

Often the population that we wish to study is very large, and it is too time-consuming or costly to conduct a census. In such a situation, we select and analyse a subset (or portion) of the population elements.

Sample: A smaller, manageable **subset** of the population for which we have data.



Census: An examination of the entire population

Note: Census usually too expensive, too time consuming, and too much effort for a large population.

The vast majority of studies use a **sample** rather than a census for logistical reason – it can be cheaper and it is easier to get data on just a sample, and you can often get “better data” by spending a lot of time with a few subjects rather than spending a little with a lot.

Examples

- With a face-to-face interviews of a sample of people, e.g., for monthly estimates of the unemployment rate
- A university graduated 8,742 students
This is too large for a census.
So, we select a sample of these graduates and learn their annual starting salaries.

1.3.3 Population of Measurements

- **Measurement** of the variable of interest for each and every population unit
For example, annual starting salaries of all graduates from last year's MBA program
- Sometimes called *observations*
- If population too large, will analyze a subset (sample)



1.3.4 Descriptive statistics (description)

Definition: refers to the methods for summarising / describing what our data shows

- For example, for a set of annual starting salaries, we want to know:
 - How much to expect
 - What is a high versus low salary
 - How much the salaries differ from each other
- If the population is small enough, could take a census and not have to sample and make any statistical inferences
- But if the population is too large, then

1.3.5 Inferential statistics (inference)

Definition: Using sample data to make **generalizations** (statements, decisions or predictions) about a population, based on data obtained from a sample of that population.

For example, use a sample of starting salaries to estimate the important aspects of the population of starting salaries

There is a **criteria on how to choose a sample:** the information contained in a sample is to accurately reflect the population under study.

- You've likely encountered descriptive statistics before – calculating averages, creating graphs, etc. These help us see patterns within our sample. (see something)
- While describing our sample is crucial, often our real goal is to go further: to use the sample to **infer** something about the **entire population**. That's where the power of inferential statistics comes in! (learn something)

Description vs inference

Example: Consider the major poll, which included 400 high school students.

- When we say 63% of respondents would vote Engineering ahead of the Social Science, we are reporting a **descriptive** statistic.
- When we use the data to answer the question “how much evidence does this study provides that the Engineering will win the Social Science? ” we are making an **inference** about the population of **all 4,000 students**, based on a sample of just 400.

Sampling introduces variation

A fundamental idea in statistical inference is that sampling induces variation – different samples will give you different data, depending on which subjects end up getting included in the sample.

Consider again the major poll example, recall that 400 high school students were sampled, and of these, 252 (63%) of them would vote for Engineering ahead of Social Science. If a different 400 high school students were sampled, would you expect to get exactly 63% voting for Engineering again?

The challenge of inference: Accounting for uncertainty

- When we use a sample to understand a larger population, we face a key challenge: **sample variation**. Different samples will naturally give slightly different results. We need to account for this inherent variability when making inferences.
- If we choose our sample **randomly** (which is often the goal!), then the **data** we collect **is also random**. This is where the power of **probability theory** comes in – it provides the framework for understanding and quantifying this randomness in our data.
- Because probability is fundamental to making sound statistical inferences, we will be building key probabilistic concepts throughout this course.

1.3.6 Random sample

A random sample is a sample selected from a population so that:

- Each population unit has the same chance of being selected as every other unit
 - Each possible sample (of the same size) has the same chance of being selected.
- For example, randomly pick two different people from a group of 15:
 - Number the people from 1 to 15; and write their numbers on 15 different slips of paper
 - Thoroughly mix the papers and randomly pick two of them
 - The numbers on the slips identifies the people for the sample

Examples



toss a coin



shuffle the cards



Roll dice



turntable

Note: the computer can also generate a random number for you.

Sample with replacement

Replace each sampled unit before picking next unit

- The unit is placed back into the population for possible reselection
- However, the same unit in the sample does not contribute new information



Sample without replacement

A sampled unit is withheld from possibly being selected again in the same sample

- Guarantees a sample of different units
 - Each sampled unit contributes different information
 - Sampling without replacement is the usual and customary sampling method.

Sometimes it is not possible to list and thus number all the units in a population. In such a situation we often select **a systematic sample**, which *approximates a random sample*.

A Systematic Sample

Randomly enter the population and systematically sample every k th unit.

Example: A Systematic Sample

- We wish to sample a population of $N = 14,327$ allergists to investigate how often they have prescribed a particular drug during the last year.
- We wish to draw a **systematic sample** of 500 allergists from this frame.
- We compute $14,327 / 500 = 28.654$, which is 28 when rounded down (**why not rounded up?**).
- Therefore, we number the first 28 allergists in the directory from 1 to 28, and we use a random number table to randomly select one of the first 28 allergists.
- Suppose we select allergist number 19. We interview allergist 19 and every 28th allergist in the frame thereafter, so we choose allergists 19, 47, 75, and so forth until we obtain our sample of 500 allergists.
- In this scheme, we must number the first 28 allergists, but we do not have to number the rest because we can “count off” every 28th allergist in the directory.

1.4 Three Case Studies that Illustrate Sampling and Statistical Inference

1. The Cell Phone Case: Estimating Cell Phone Costs
2. The Marketing Research Case: Rating a New Bottle Design
3. The Car Mileage Case: Estimating Mileage

Example 1.1: The Cell Phone Case: Estimating Cell Phone Costs

- A Cellular Management Services company claims that if its cellular cost per minute is over 18 cents per minute, the bank will benefit from automated cellular management of its calling plans
- A bank is considering using a Cellular Management Services company to manage their cellular resources
- The bank has over 4,000 employees. Random sample of 100 employees on 500-minute plan.



TABLE 1.4 A Sample of Cellular Usages (in minutes) for 100 Randomly Selected Employees

 CellUse

75	485	37	547	753	93	897	694	797	477
654	578	504	670	490	225	509	247	597	173
496	553	0	198	507	157	672	296	774	479
0	822	705	814	20	513	546	801	721	273
879	433	420	521	648	41	528	359	367	948
511	704	535	585	341	530	216	512	491	0
542	562	49	505	461	496	241	624	885	259
571	338	503	529	737	444	372	555	290	830
719	120	468	730	853	18	479	144	24	513
482	683	212	418	399	376	323	173	669	611

The units of
the number
in the table:
minute

Many overages and underage.

The result

If we add together the usages, we find that the 100 employees used a total of 46,625 minutes. Furthermore, the total cellular cost incurred by the 100 employees is found to be \$9,317 (this total includes base costs, overage costs, long distance, and roaming). This works out to an average of $\$9,317 / 46,625 = 0.1998$, or 19.98 cents per minute. Because this average cellular cost per minute exceeds 18 cents per minute, the bank will hire the cellular management service to manage its calling plans.

Example 1.2: The Marketing Research Case: Rating New Bottle Design

- A brand group designed a new bottle.
- The group interviews consumers reactions to a new bottle design.
- Using “mall intercept method.”
- Sample size of 60.



FIGURE 1.2 The Bottle Design Survey Instrument

Please circle the response that most accurately describes whether you agree or disagree with each statement about the bottle you have examined.

Statement	Strongly Disagree					Strongly Agree	
The size of this bottle is convenient.	1	2	3	4	5	6	7
The contoured shape of this bottle is easy to handle.	1	2	3	4	5	6	7
The label on this bottle is easy to read.	1	2	3	4	5	6	7
This bottle is easy to open.	1	2	3	4	5	6	7
Based on its overall appeal, I like this bottle design.	1	2	3	4	5	6	7

Experience has shown that the smallest acceptable bottle image composite score for a successful bottle design is 25.

TABLE 1.5 A Sample of Bottle Design Ratings (Composite Scores for a Sample of 60 Shoppers)

34	33	33	29	26	33	28	25	32	33
32	25	27	33	22	27	32	33	32	29
24	30	20	34	31	32	30	35	33	31
32	28	30	31	31	33	29	27	34	31
31	28	33	31	32	28	26	29	32	34
32	30	34	32	30	30	32	31	29	33

Most consumers would rate the new bottle design between 20 and 35. Furthermore, 57 of the 60 composite scores are at least 25. Therefore, we might estimate that a proportion of $57 / 60 = 0.95$ (that is, 95 percent) of all consumers would give the bottle design a composite score of at least 25.

Definitions:

- **Process:** a sequence of operations that takes inputs and turns them into outputs
- **Finite population:** a population of limited size
- **Infinite population:** a population of unlimited size

Sampling a Process

Process

A sequence of operations that takes *inputs* (labor, raw materials, methods, machines, and so on) and turns them into *outputs* (products, services, and the like)



Example 1.3: The Car Mileage Case: Estimating Mileage

- **Auto fuel economy**: Study of tax credit offered by the federal government for improving fuel economy.
- **Sampling a process**: Automaker has introduced **a new model** and wishes to demonstrate it qualifies for the tax credit. In order to study the population of all cars of this type that will or could potentially be produced, the automaker will choose **a sample of 50 of these cars**.

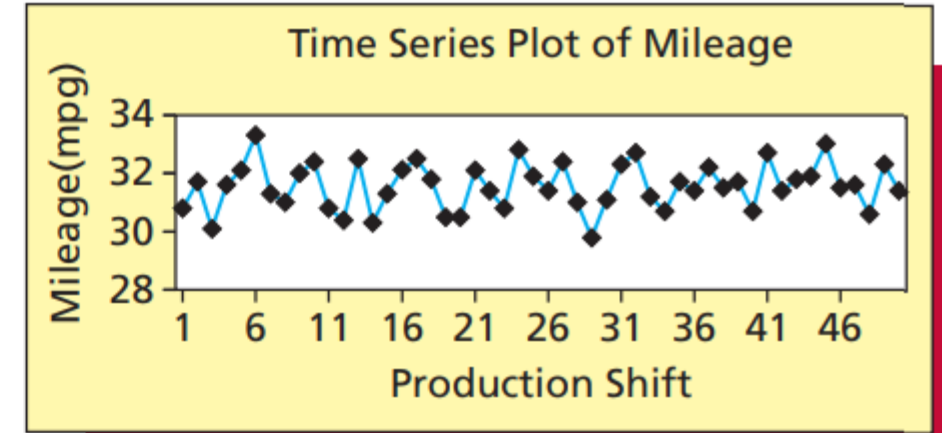


TABLE 1.6 A Sample of 50 Mileages 

30.8	30.8	32.1	32.3	32.7
31.7	30.4	31.4	32.7	31.4
30.1	32.5	30.8	31.2	31.8
31.6	30.3	32.8	30.7	31.9
32.1	31.3	31.9	31.7	33.0
33.3	32.1	31.4	31.4	31.5
31.3	32.5	32.4	32.2	31.6
31.0	31.8	31.0	31.5	30.6
32.0	30.5	29.8	31.7	32.3
32.4	30.5	31.1	30.7	31.4

Note: Time order is given by reading down the columns from left to right.

FIGURE 1.3 A Time Series Plot of the 50 Mileages



Conclusion

Although the mileages vary over time, they do not seem to vary in any unusual way.

Verifies consistent car mileages over time.

Thus, we can regard the 50 mileages as **an approximately random sample** that **can** be used to **make** statistical **inferences** about the population of all possible midsize car mileages.

Exercises for Sections 1.3 and 1.4 (if time permits)

1.8 Define a population. Give an example of a population.

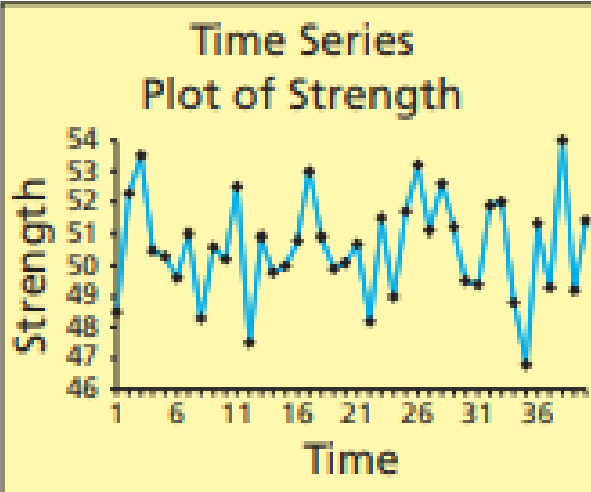
1.9 Explain the difference between a census and a sample.

1.14 The trash bag case – the strength

- A company has developed an improved 30-gallon bag.
- The company wishes to evaluate the strength of this bag.
- The company has decided to select a sample of 40 of the new trash bags.
- Each bag is subjected to a breaking strength test.
- The 40 breaking strengths obtained are given in Table 1.9.
- Estimate limits between which the breaking strengths of most trash bags would fall.

TABLE 1.9
Trash Bag
Breaking Strengths
DS TrashBag

48.5	50.7
52.3	48.2
53.5	51.5
50.5	49.0
50.3	51.7
49.6	53.2
51.0	51.1
48.3	52.6
50.6	51.2
50.2	49.5
52.5	49.4
47.5	51.9
50.9	52.0
49.8	48.8
50.0	46.8
50.8	51.3
53.0	49.3
50.9	54.0
49.9	49.2
50.1	51.4



Thank you!